Dynamic Pathfinding Agent

AI-2002 Assignment 02 - Question 6

Student: 22F3098 | BS(AI)4A1

1. Objective

The objective of this project is to develop a Dynamic Pathfinding Agent capable of navigating a grid-based environment using informed search algorithms. The system handles both static and dynamic obstacles and performs real-time re-planning when necessary.

2. Environment Design

The environment is implemented using Python and Pygame. It supports dynamic grid sizing, fixed start and goal nodes, random obstacle generation, manual obstacle placement using mouse clicks, and dynamic obstacle spawning during execution.

3. Algorithms Implemented

A* Search uses f(n) = g(n) + h(n) and guarantees an optimal path when the heuristic is admissible and consistent. Greedy Best First Search (GBFS) uses f(n) = h(n) and selects nodes purely based on heuristic value without considering path cost.

4. Heuristic Functions

Two heuristics are implemented: Manhattan Distance (|x1-x2| + |y1-y2|) suitable for 4-directional grid movement, and Euclidean Distance (sqrt((x1-x2)^2 + (y1-y2)^2)) which estimates straight-line distance.

5. Dynamic Mode & Re-planning

In dynamic mode, obstacles spawn randomly while the agent is moving. If a new obstacle blocks the current path, the agent detects the obstruction and recalculates the path from its current position instead of restarting from the beginning.

6. Metrics

The system calculates Nodes Visited (expanded nodes count), Path Cost (length of final path), and Execution Time in milliseconds to compare algorithm performance.

7. Experimental Analysis

In best-case scenarios with fewer obstacles, both A* and GBFS perform efficiently. In worst-case maze-like environments, GBFS may explore misleading paths while A* consistently finds the optimal solution.

8. Comparison

A* guarantees optimality but may consume more memory. GBFS is faster in simple maps but does not guarantee the shortest path.

9. Conclusion

The Dynamic Pathfinding Agent successfully demonstrates the behavior of informed search algorithms in both static and dynamic environments. A* proves to be reliable and optimal, while GBFS offers faster but less accurate solutions.

Attached Screenshots (Original Submission)

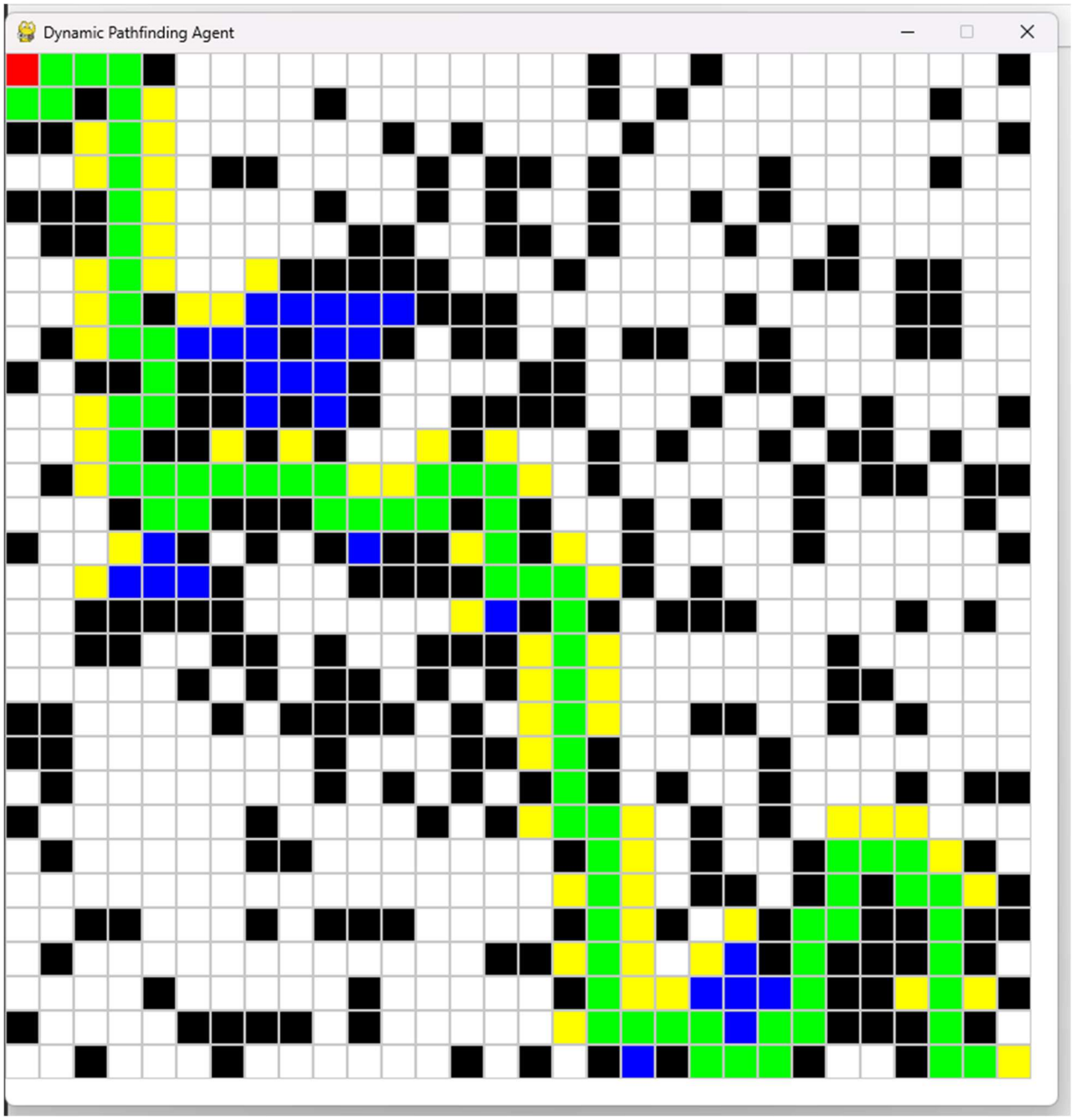
22F3098

BS(AI)4A1

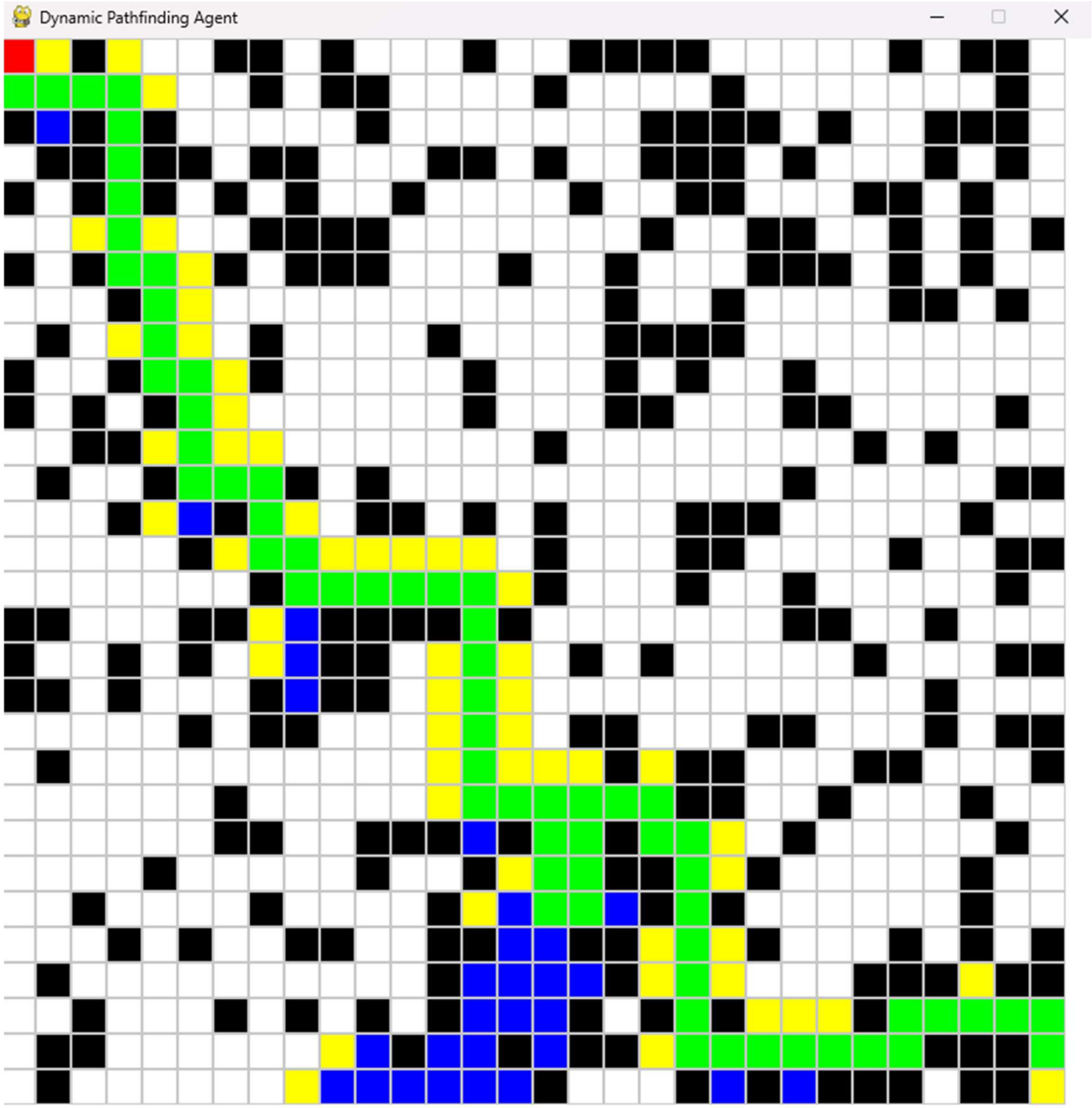
Q6:

A*

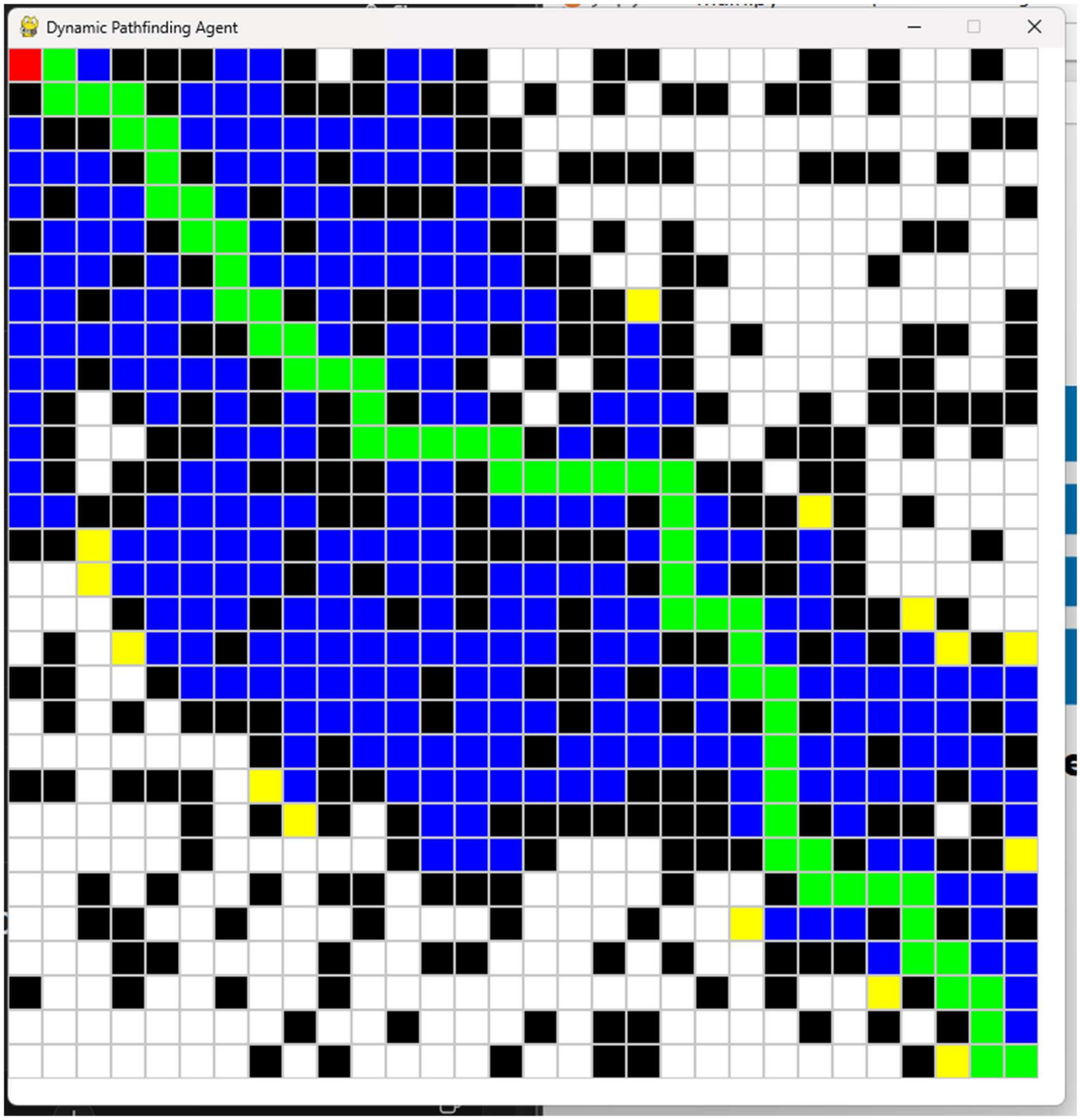
GBFS:



MANHATTAN:



EUCLIDEAN


Dynamic Pathfinding Agent